Tree Borrows

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Rust's type system enables powerful optimizations

```
fn write_both(x: &mut i32, y: &mut i32) -> i32 {

  *x = 13;

  *y = 20;

  *x

}
```

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fn write_both(x: &mut i32, y: &mut i32) -> i32 {
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mutable thus disjoint

*x is unchanged

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  *y = 20;
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```

mutable thus disjoint

*x has known value

*x is unchanged

always returns 13

Rust's type system enables powerful optimizations

```
fn write_both(x: &mut i32, y: &mut i32) -> i32 {
  *x = 13;
  *y = 20;
  13 // formerly *x: one fewer load from memory
}
```

Type-level guarantees for references

aliasing
&
mutability

`&mut` → mutation, no aliasing

`&` → aliasing, no mutation

Escape hatch: `unsafe`

Can use **unchecked operations** to do **low-level manipulations**

```
unsafe {
  // Code within this block can effectively
  // bypass some parts of the typechecker.
  ...
}
```

Within `unsafe` it is **the programmer's responsibility** to check

- that pointers are non-null
- that memory is initialized
- absence of data races
- …

violations trigger UB

Why have Undefined Behavior (UB) in a language?

Code that contains UB can have **any behavior**.

Relevance for optimizations:

allowed to change the behavior of programs that contain UB.

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Code that contains UB can have **any behavior**.

Relevance for optimizations:
allowed to change the behavior of programs that contain UB.

Too little UB $\rightarrow$ weak optimizations

Too much UB $\rightarrow$ hard to write correct programs

What if `unsafe` code is misused ?

```rust
fn write_both(x: &mut i32, y: &mut i32) -> i32 {
  *x = 13;
  *y = 20;
  *x
}

fn main() {
  let mut root = 42;
  let ptr = &raw mut root;
  let x = unsafe { &mut *ptr };
  let y = unsafe { &mut *ptr };
  println!("{}", write_both(x, y)); // prints 20
}
```

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Not the compiler's responsibility

Within `unsafe` it is **the programmer's responsibility** to check

- that pointers are non-null
- that memory is initialized
- absence of data races
- compliance with aliasing rules NEW!

violations trigger UB

Tree Borrows (TB): defines those aliasing rules

Not the compiler's responsibility

Within `unsafe` it is **the programmer's responsibility** to check

- that pointers are non-null
- that memory is initialized

Sounds familiar?

Stacked Borrows has the same purpose,

Tree Borrows is its successor.

Tree Borrows (TB): defines those aliasing rules

Stacked Borrows (SB)

In safe Rust, the Borrow Checker makes borrows well-bracketed.

Stacked Borrows extends the well-bracketedness to `unsafe`.

```
let mut root = 42;
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```

root

- new stack at root

Desired outcome: UB

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- ✓ root is at the top

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```

ptr
root

- ✓ `root` is at the top
- push `ptr`

Desired outcome: UB

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x
ptr
root

- ✓ ptr is at the top
- push x

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x
ptr
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- pop until `ptr` is at the top

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root

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y
ptr
root

- pop until `ptr` is at the top
- push y

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```

y
ptr
root

- search for x

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```

UB!

y
ptr
root

Can’t use x if it is not in the stack

Desired outcome: UB

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$\to$ included in many projects' CI
$\to$ several bugs detected (e.g. in stdlib)

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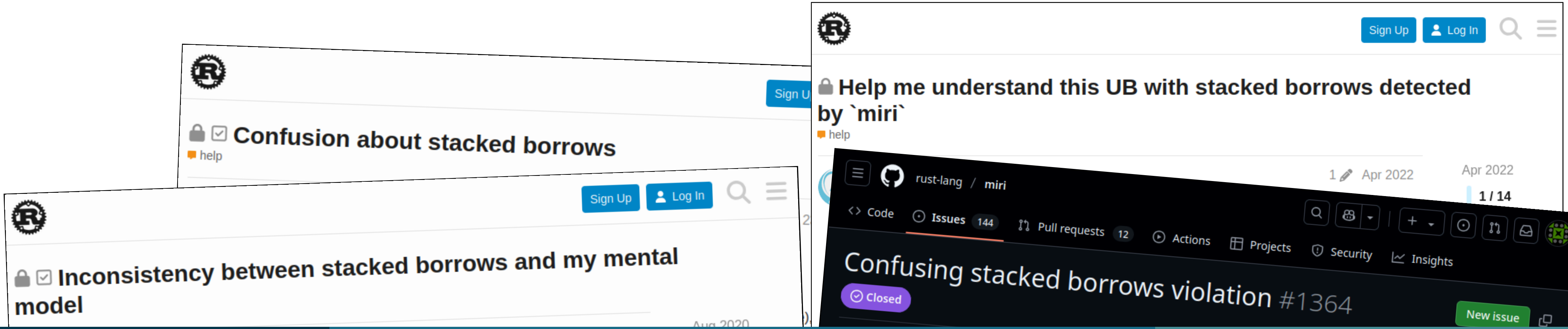

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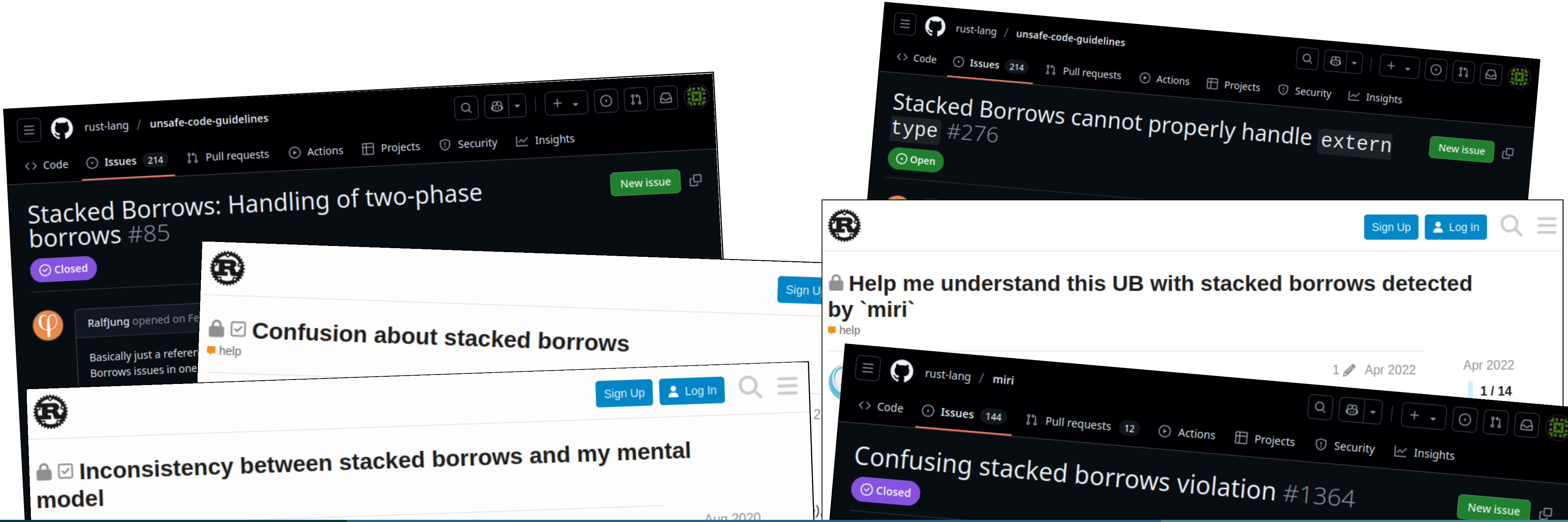
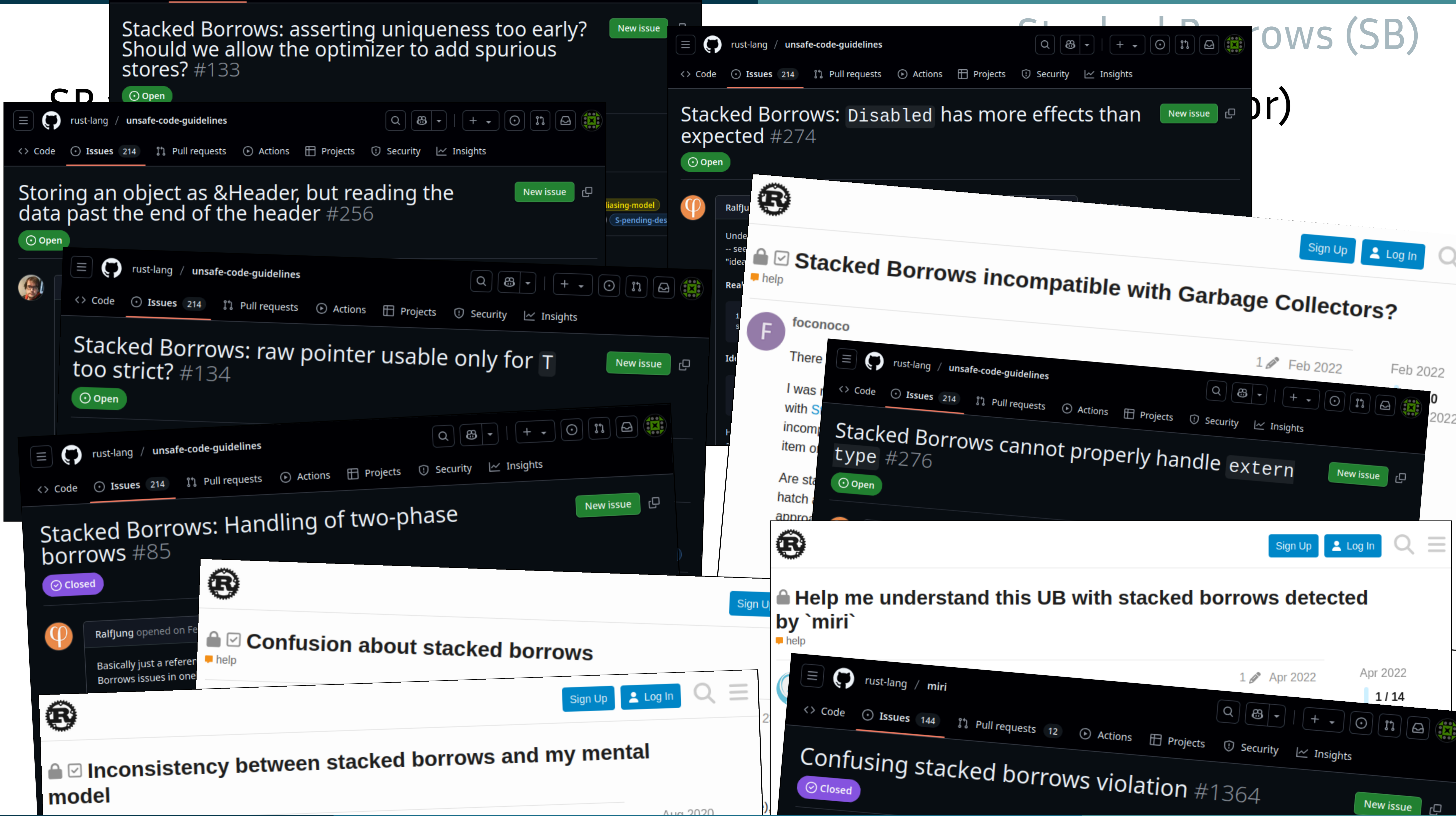
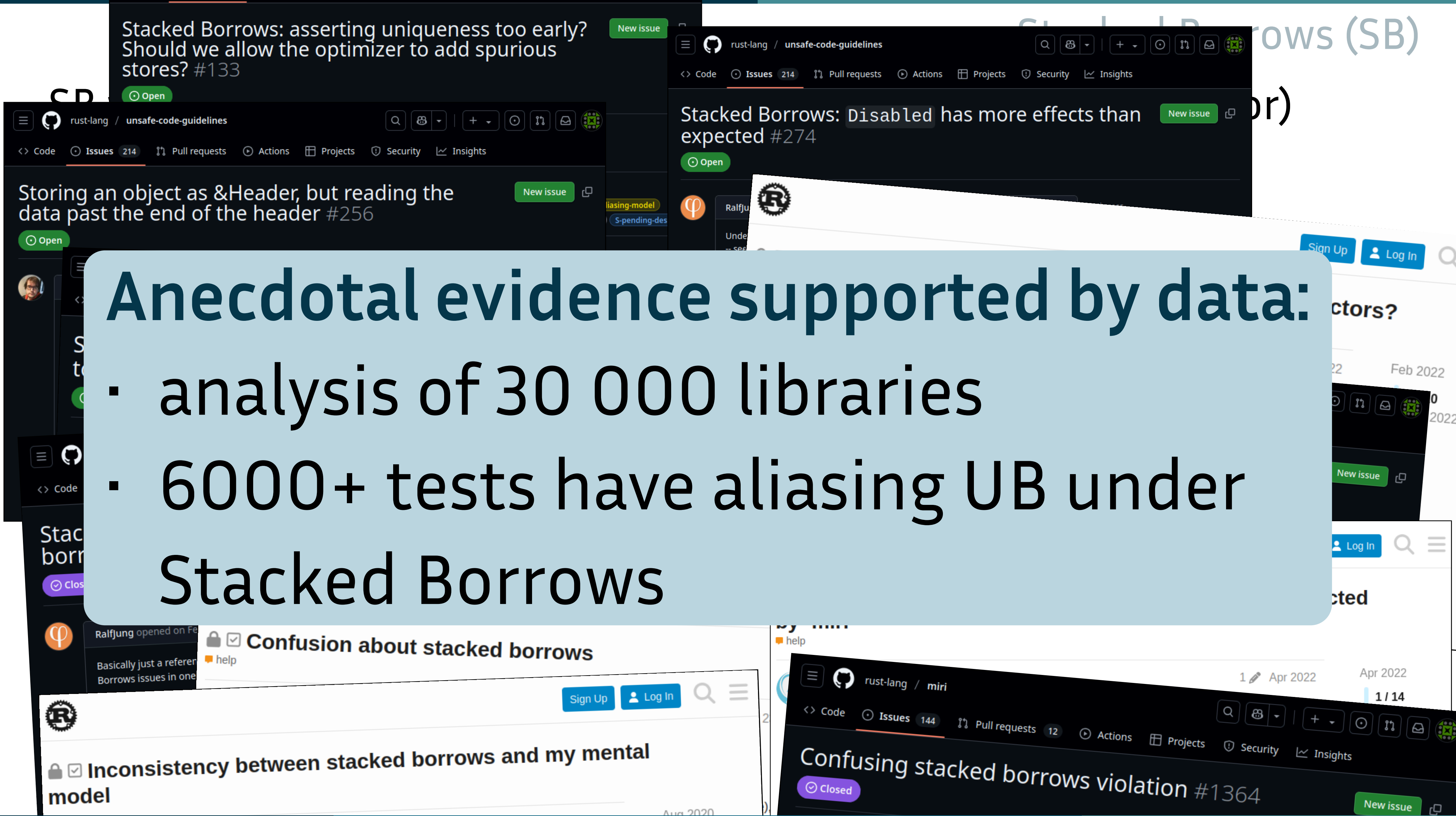



Stacked Borrows: asserting uniqueness too early? Should we allow the optimizer to add spurious stores? #133
Open
Stacked Borrows: Disabled has more effects than expected #274
Open
Storing an object as &Header, but reading the data past the end of the header #256
Open
Stacked Borrows: raw pointer usable only for T too strict? #134
Open
Stacked Borrows: Handling of two-phase borrows #85
Closed
Stacked Borrows incompatible with Garbage Collectors?
help
foconoco
Stacked Borrows cannot properly handle extern type #276
Open
Confusion about stacked borrows
help
Help me understand this UB with stacked borrows detected by `miri`
help
Inconsistency between stacked borrows and my mental model
Confusing stacked borrows violation #1364
Closed



Anecdotal evidence supported by data:

- analysis of 30 000 libraries
- 6000+ tests have aliasing UB under Stacked Borrows

Tree Borrows allows much more code

Tree Borrows uses a **tree** instead of a stack to track borrows

Out of 30 000 most downloaded libraries,
$> 50\%$ **fewer tests** with aliasing UB when using Tree Borrows

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fixes known technical limitations of SB,
incl. 2-phase borrows, extern types, **pointer offsets**

From Stacks to Trees

```rust
let mut root = vec!['a','b','c'];
let x0 = &raw mut root[0];
let x2 = &raw mut root[2];
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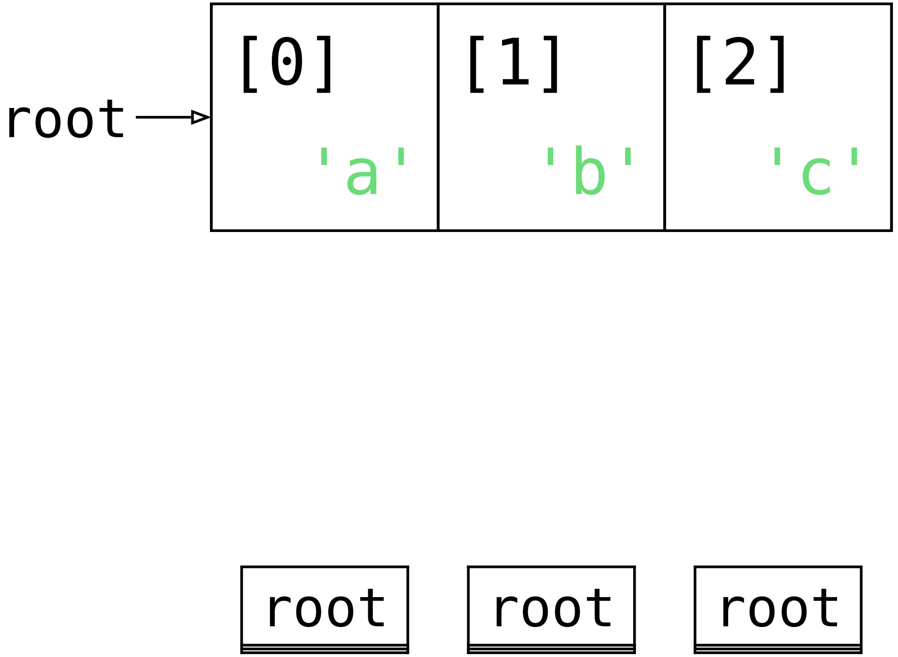

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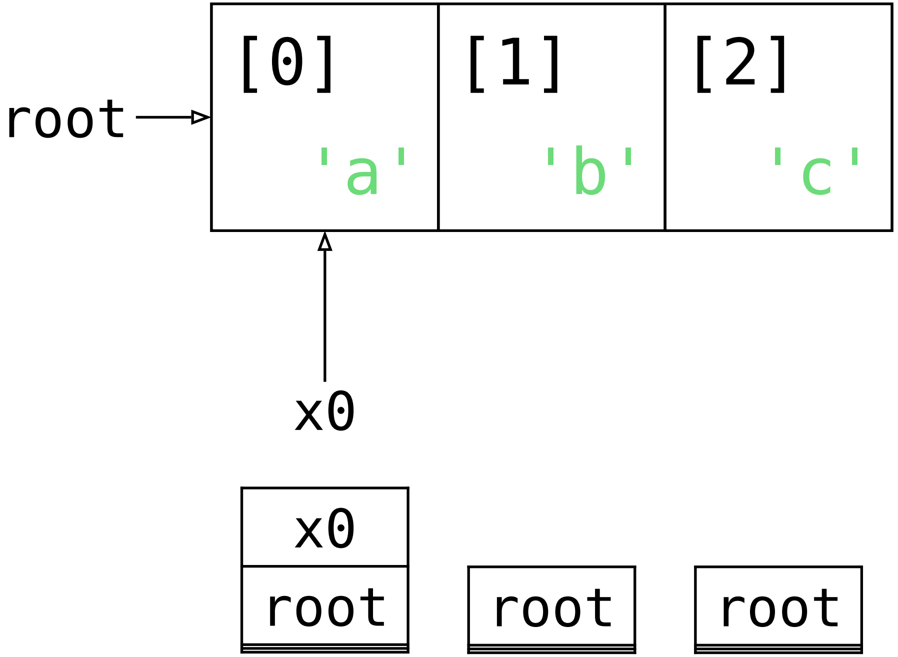

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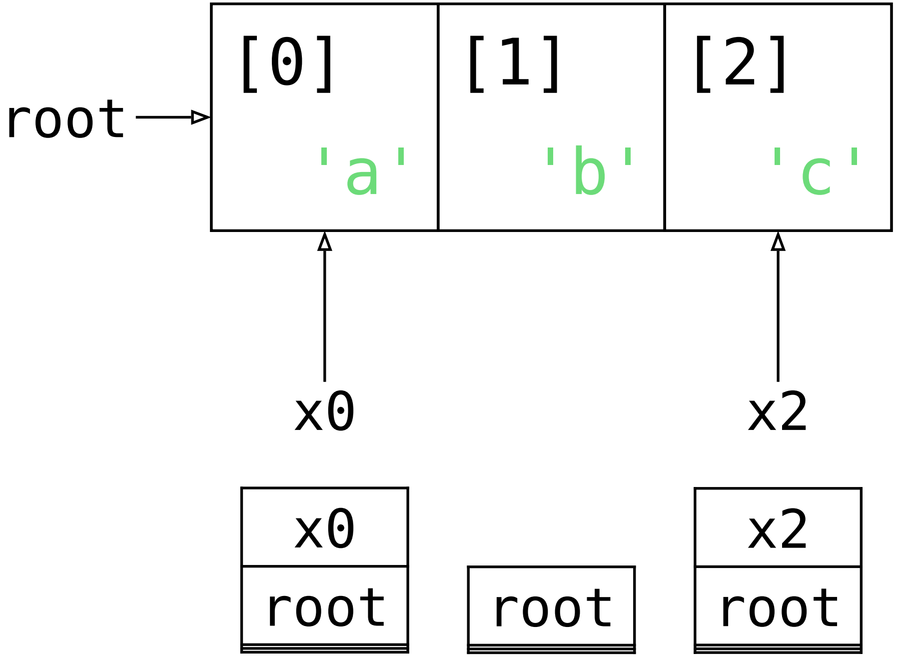

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let mut root = vec!['a','b','c'];
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// Scenario 1
unsafe { *x0.add(1) = 'z'; }
```

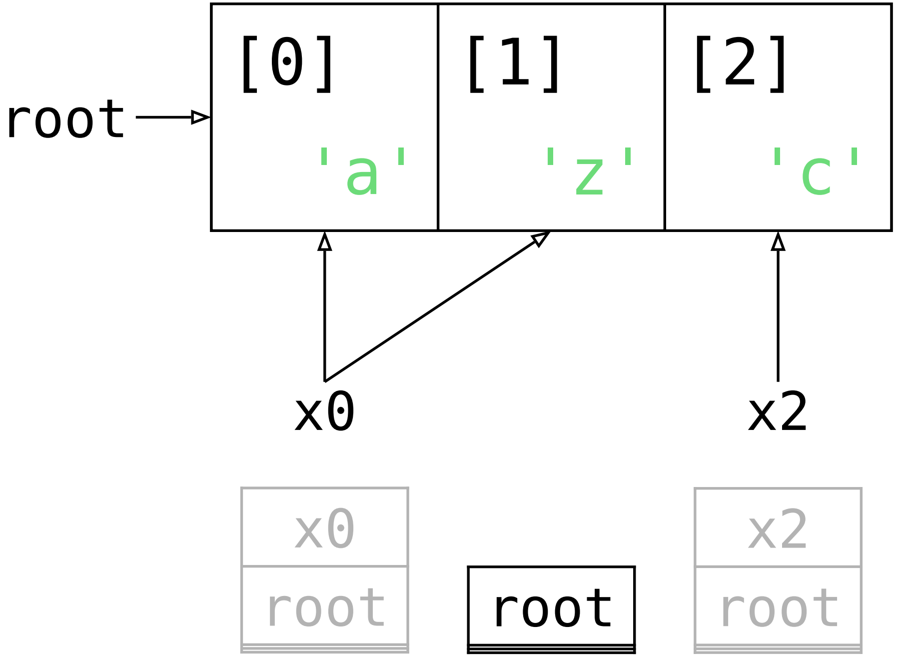

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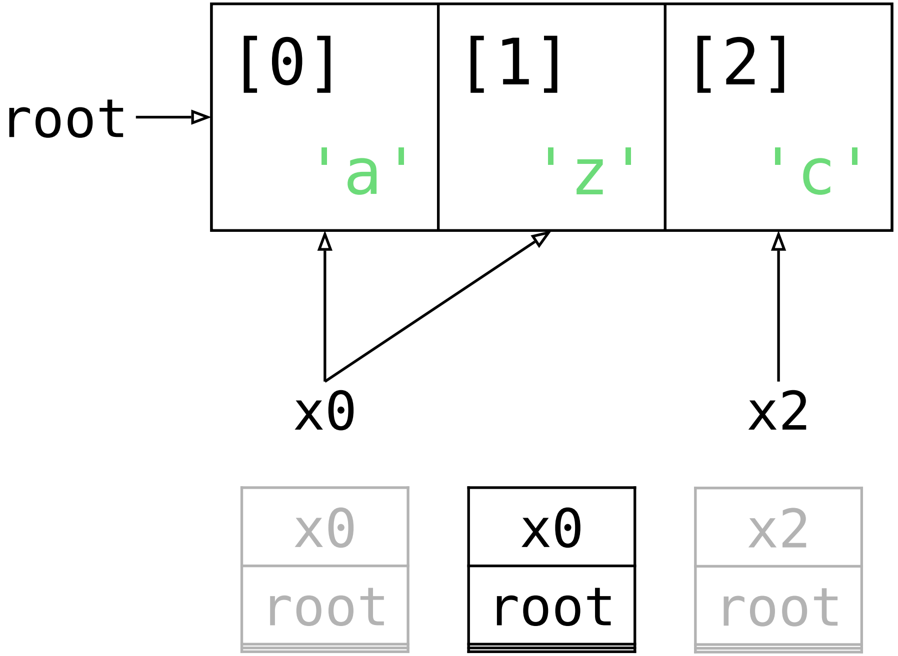

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```
let mut root = vec!['a','b','c'];
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// Scenario 2
unsafe { *x2.sub(1) = 'z'; }
```

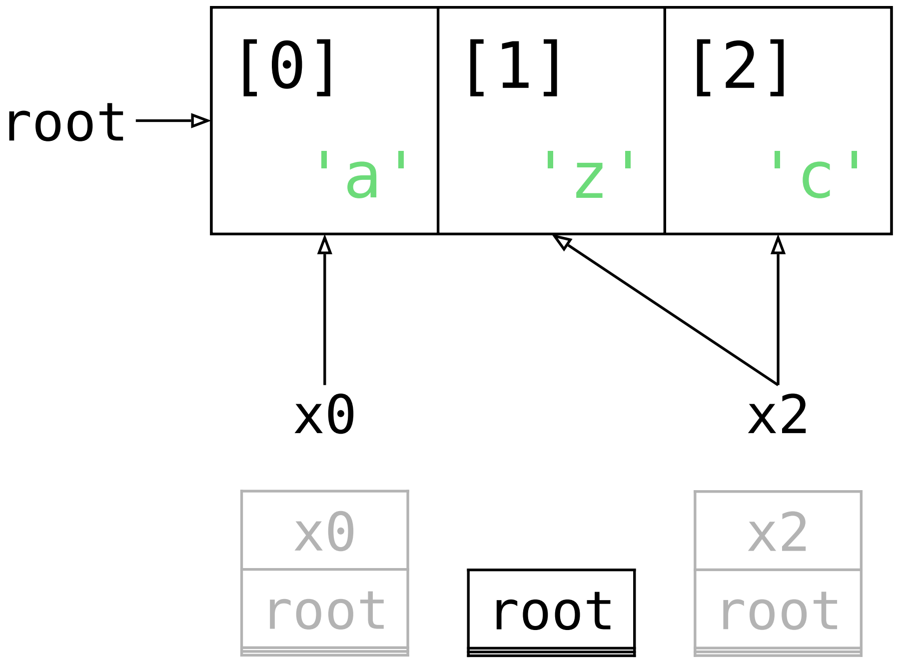

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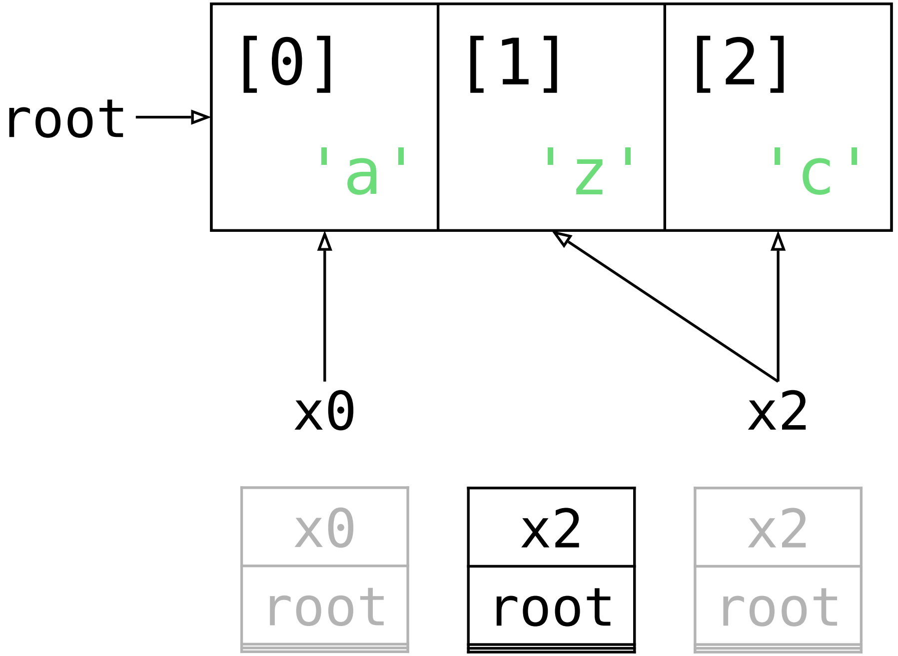

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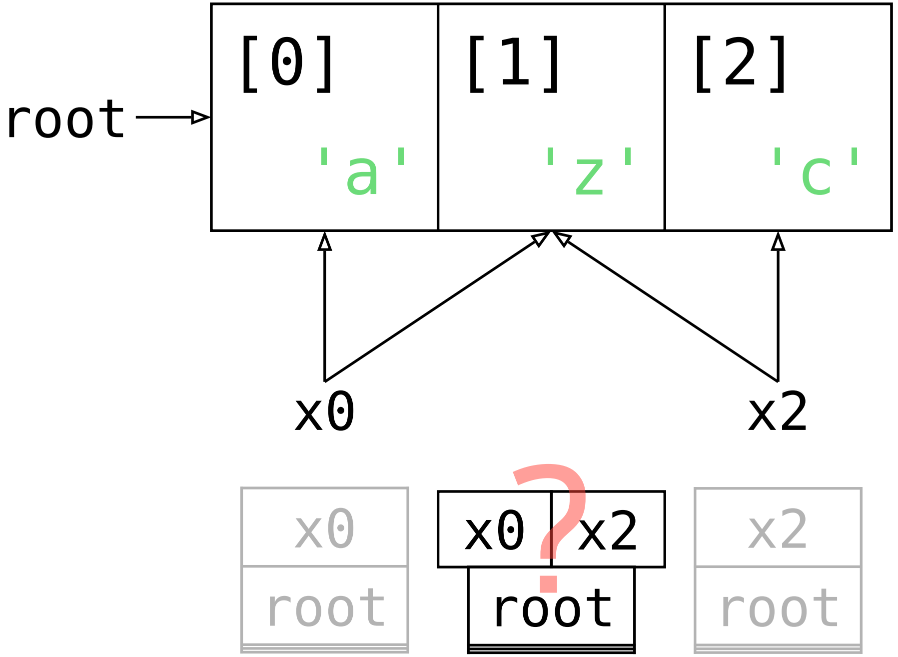
// Scenario 1 or 2
unsafe { *??? = 'z'; }
```



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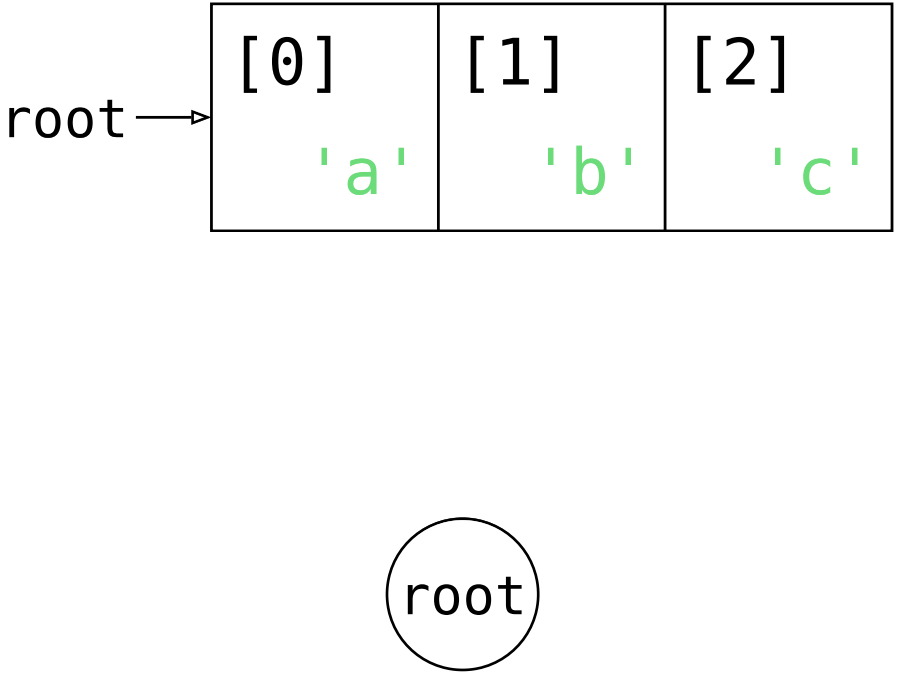

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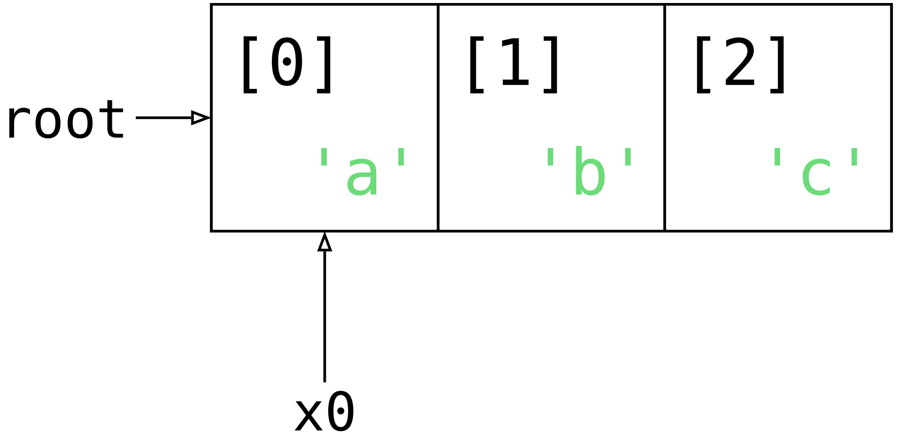

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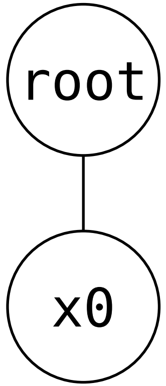

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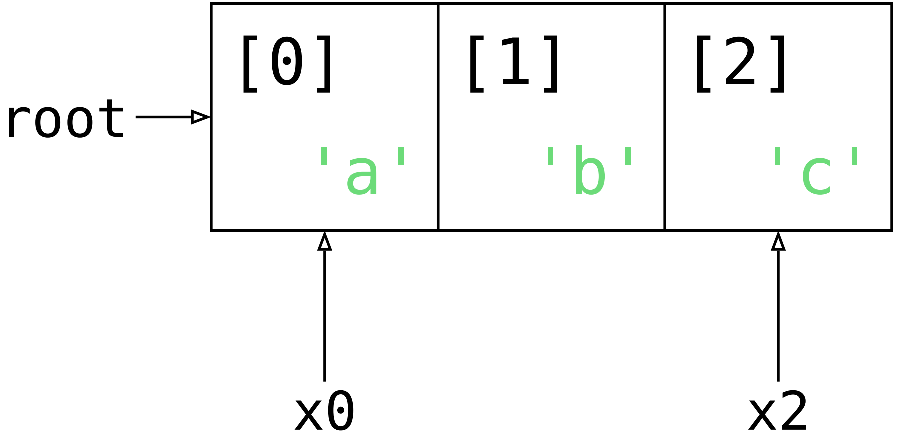

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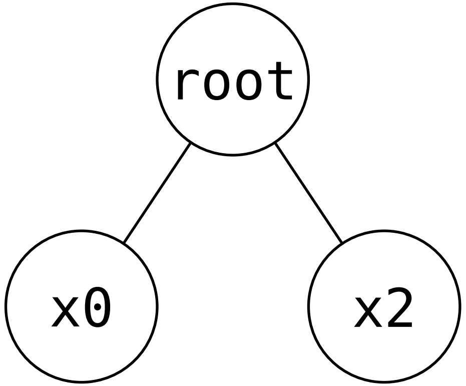

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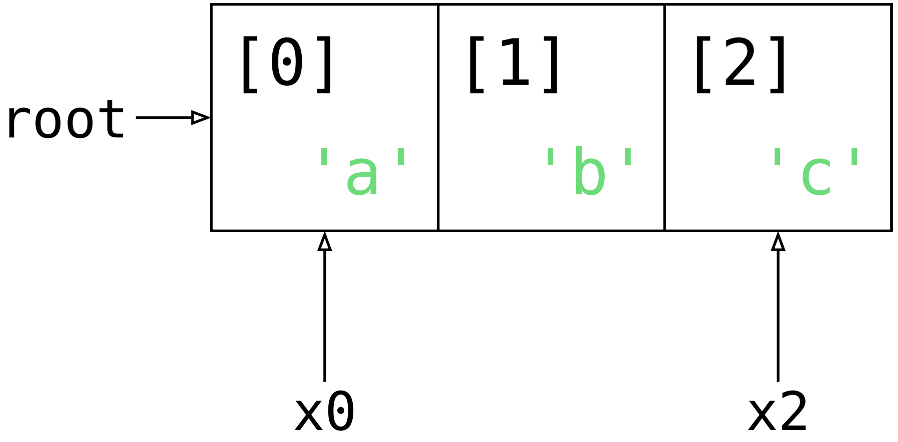

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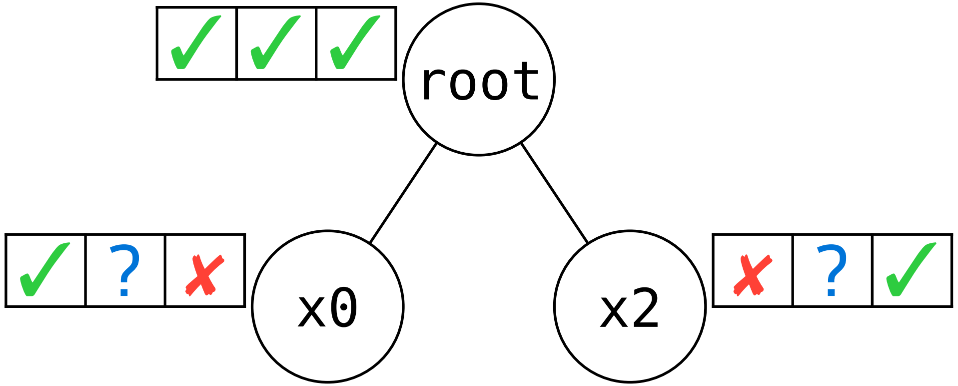

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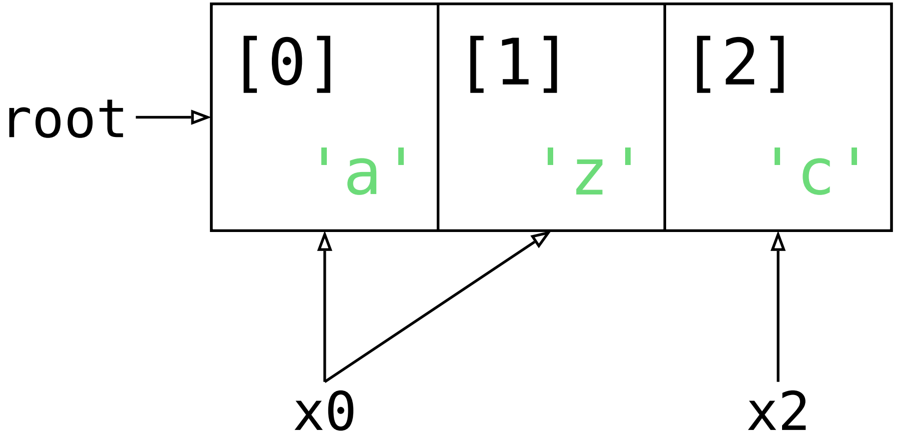

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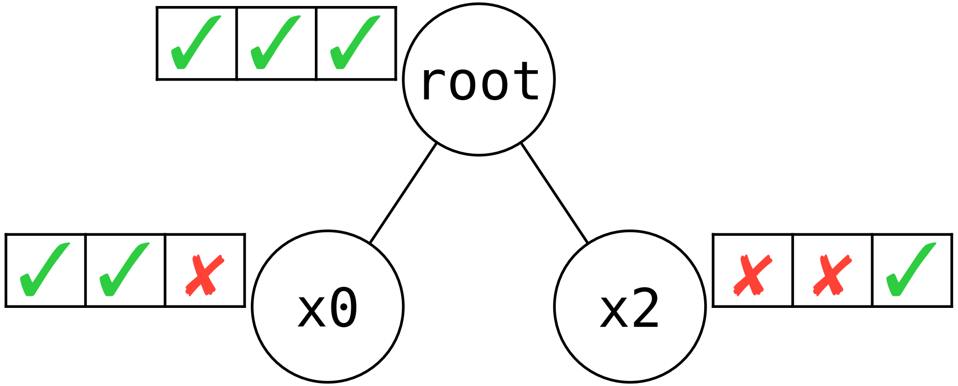

```
let mut root = vec!['a','b','c'];
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// Scenario 2
unsafe { *x2.sub(1) = 'z'; }
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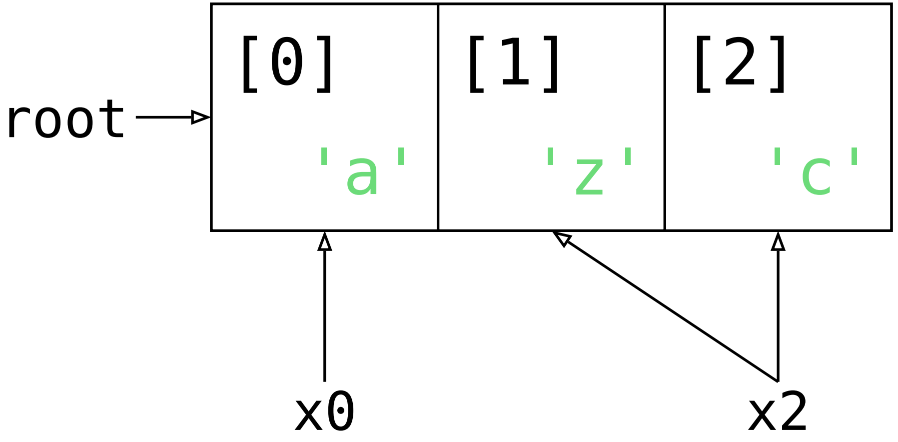

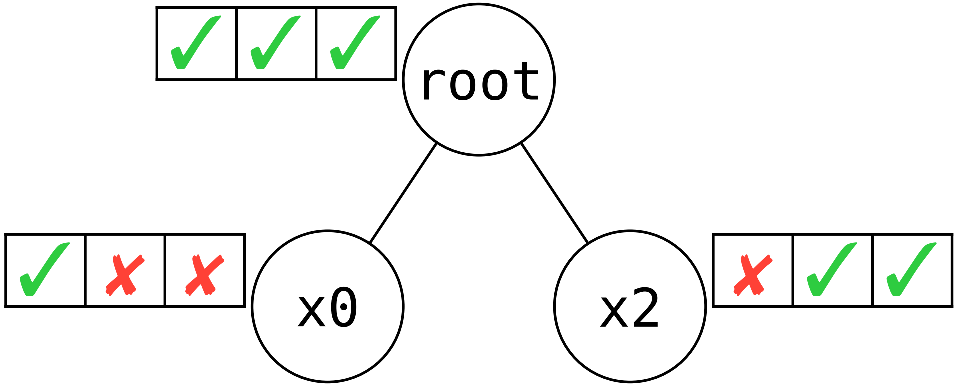

A second look at the motivating example


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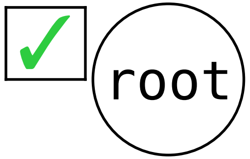

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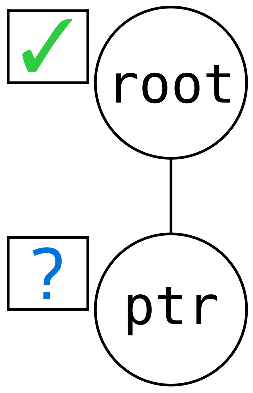

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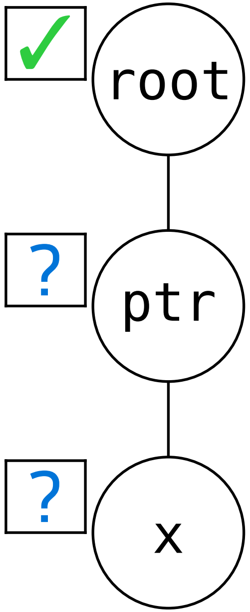

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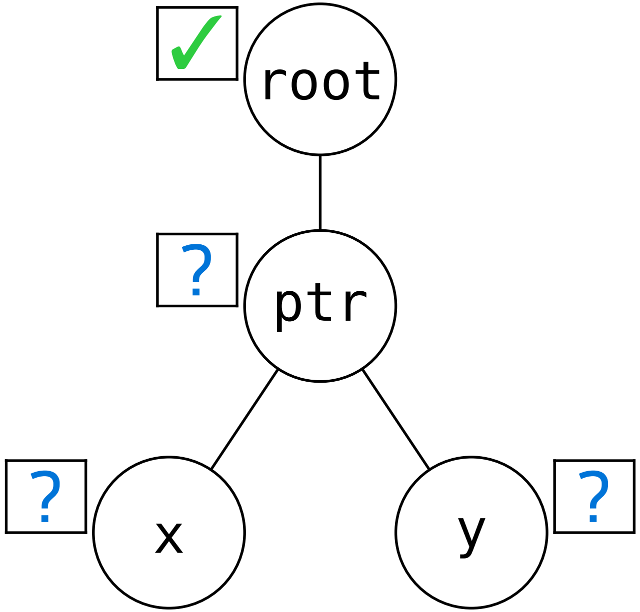

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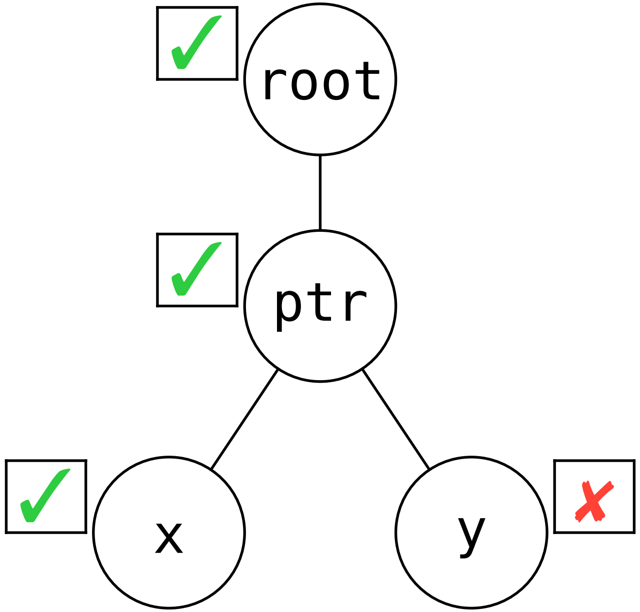

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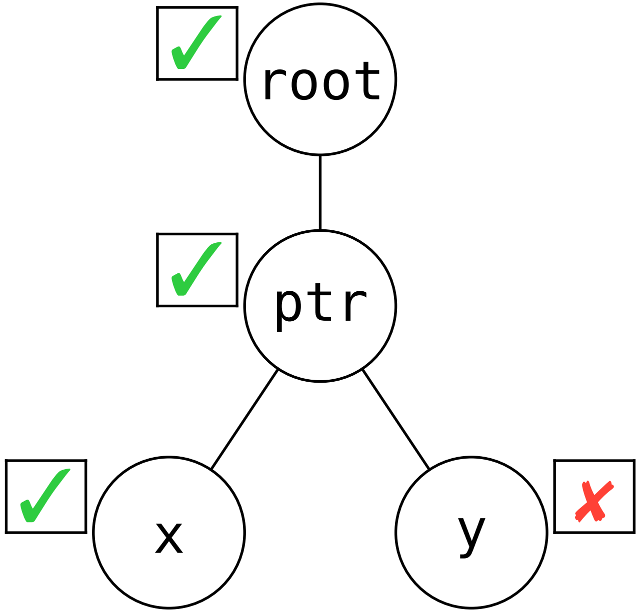
UB!



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Evaluation

TB should enable desired optimizations

i.e. have enough UB to rule out problematic patterns

- formalized in Rocq (+Simuliris)
- a selection of optimizations proven
  - ✓ delete read through `&mut` or `&`
  - ✓ insert read through `&` in function
  - ✓ move read down for `&mut` or `&` in function

  …

It should be possible to write `unsafe` code free of UB

i.e. UB should be predictable and not too common

50% fewer tests have aliasing UB according to Tree Borrows

Only 31 ($< 0.01\%$) tests are regressions, all easily fixable.

(Out of 30 000 libraries, 400 000+ working tests)

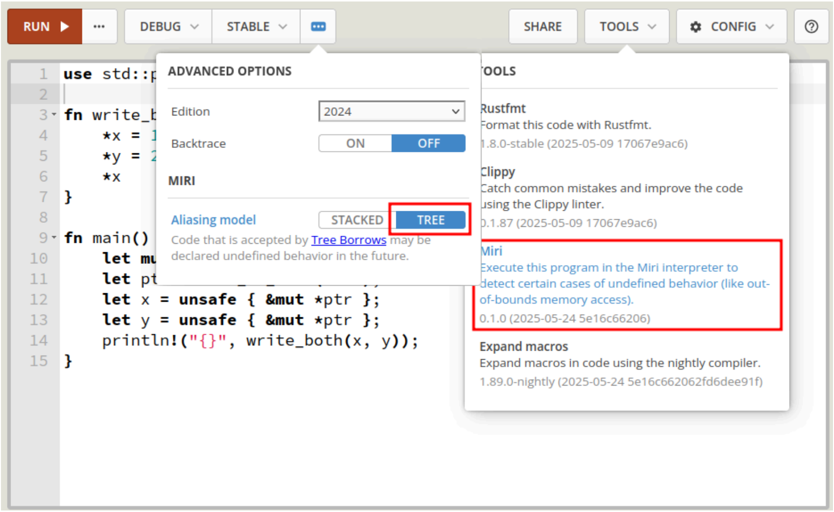
"Tree Borrows accepts more real-world programs that call foreign functions than Stacked Borrows due to differences in handling pointer arithmetic."

A Study of Undefined Behavior Across Foreign Function Boundaries in Rust Libraries, by I. McCormack, J. Sunshine, J. Aldrich @ ICSE'25

Conclusion

Try it out: supported by Miri

Also on the Rust Playground (`play.rust-lang.org`)



Learn more:

`plf.inf.ethz.ch/research/pldi25-tree-borrows.html`

Includes e.g. handling of raw pointers and interior mutability.

Postdoc positions available

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